3. Module 1: Getting Familiar

This module teaches students to employ AI to train, generate, analyse, and visualise data for specific design purposes {Module 1-Obj}. The module is organised into three sub-modules to get familiar with a specific discipline: Machine Learning (ML), Generative AI (Gen AI) and Computer Vision (CV). Each sub-module is intended to provide basic literacy, guided tutorials of tools, hands-on experience and documentation practices of results. To promote the development of personal skills, students are suggested to work individually.

Teachers can arrange the duration of each activity based on the level of in-depth study required [example]. Sub-modules are not strictly related to each other. The application of all of them is not mandatory, teachers can make a selection based on needs and interests. The structure of the sub-modules is not designed based on the tool mentioned in the guideline. For a more complete list of tools, refer to the Designing With framework [link].

Educational Objective {MOD1-Obj} :

Students will learn to employ AI to train, generate, analyze and visualize data for specific design purposes.

Sub-mod 2: Getting familiar w/ GenAI	The cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{Act.1}				
(b) Conceptual Knowledge		{Act.2} {Act.3}				
(c) Procedural Knowledge		{Act.4} {Act.4.1, 4.2, 4.3}	{MOD1-Obj} {Act.5}			
(d) Meta Knowledge				{Act.6}	{Act.7}	

{Act.1} Activities intended to provide students with vocabulary knowledge of AI;
{Act.2} Activities intended to provide students with a basic functionality of AI;
{Act.3} Activities intended to provide students with practical examples of AI practices applied to design;
{Act.4} Activities intended to explain the procedure of train, generate, analyse, and visualise data with AI;
{Act.4.1} Activities intended to introduce students in familiarising, collecting or creating data;
{Act.4.2} Activities intended to set or aligning with a design goal;
{Act.4.3} Activities intended to provide a step-by-step tool tutorial, such as RunwayML, DallE (genAI), Google Vision API (CV);
{Act.5} Activities intended to allow students to individually train, generate, analyse, and visualise data with AI;
{Act.6} Activities intended to provide students with a structured template for process and results documentation;
{Act.7} Activities intended to make students explaining and questioning the process.

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3.1 Sub-module 1: Getting Familiar with Machine Learning (ML)

This sub-module teaches students to train simplified ML models to create design artefacts {ML-Obj}. Students develop procedural knowledge through practical guided activities (e.g. RunwayML step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating a logo). Students are first provided with a basic literacy of ML including technical vocabulary {ML-Act1} and primary functionalities {ML-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of ML tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of ML within the design practice {ML-Act3}.

At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial {ML-Act4.3} on how to use an ML tool (e.g. runway) starting from setting a design goal {ML-Act4.1} and creating their dataset {ML-Act4.2}. Then, to familiarise themselves with the procedure, students individually apply what they learned by training an ML model aligned with the design goal {ML-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results {ML-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {ML-Act7}.

Educational Objective {ML-Obj} :

Students will learn to train a Machine Learning model to create design artifacts.

Sub-module 1: Getting familiar w/ ML	The cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{ML-Act.1}				
(b) Conceptual Knowledge		{ML-Act.2} {ML-Act.3}				
(c) Procedural Knowledge		{ML-Act.4} {ML-Act.4.1, 4.2, 4.3}	{ML-Obj} {ML-Act.5}			
(d) Meta Knowledge				{ML-Act.6}	{ML-Act.7}	

{ML-Act1} Activities intended to provide students with vocabulary knowledge of ML;
{ML-Act2} Activities intended to provide students with a basic functionality of ML;
{ML-Act3} Activities intended to provide students with practical examples of machine learning applied to design;
{ML-Act4} Activities intended to explain the procedure of training a simplified machine learning model;
{ML-Act4.1} Activities intended to set or aligning with a design goal;
{ML-Act4.2} Activities intended to introduce students in familiarising, collecting or creating data;
{ML-Act4.3} Activities intended to provide a step-by-step tool tutorial, such as RunwayML;
{ML-Act5} Activities intended to allow students to individually train a simplified machine learning model for a design goal;
{ML-Act6} Activities intended to provide students with a structured template for process and results documentation;
{ML-Act7} Activities intended to make students explaining and questioning the process.

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3.2 Sub-module 1: Getting Familiar with Generative AI (GenAI)

This sub-module teaches students to produce content with generative AI tools to create design artefacts [GenAI-Obj]. Students develop procedural knowledge through practical guided activities (e.g. Midjourney step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating an original shape for a lamp inspired by nature). Students are first provided with a basic literacy of AI including technical vocabulary (GenAI-Act1) and primary functionalities (GenAI-Act2). This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the discipline and their interrelationship to explain the functionality of AI tools. Once basic literacy is introduced, specific case studies are provided to contextualise the application of AI within the design practice (GenAI-Act3).

At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial (ML-Act4.3) on how to use an ML tool (e.g. runway) starting from setting a design goal (ML-Act4.1) and creating their dataset (ML-Act4.2). Then, to familiarise themselves with the procedure, students individually apply what they learned by training an ML model aligned with the design goal (ML-Act5). Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results (ML-Act6). This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others (ML-Act7).

Educational Objective {GenAI-Obj} :

Students will learn to **generate contents with GenAI tools** to **create design artifacts.**

Sub-module 1: Getting familiar w/ GenAI	The cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{GenAI-Act.1}				
(b) Conceptual Knowledge		{GenAI-Act.2} {GenAI-Act.3}				
(c) Procedural Knowledge		{GenAI-Act.4} {GenAI-Act.4.1, 4.2, 4.3}	{GenAI-Obj} {GenAI-Act.5}			
(d) Meta Knowledge				{GenAI-Act.6}	{GenAI-Act.7}	

{Act.1} Activities intended to provide students with vocabulary knowledge of GenAI;
{Act.2} Activities intended to provide students with a basic functionality of GenAI;
{Act.3} Activities intended to provide students with practical examples of GenAI applied to design;
{Act.4} Activities intended to explain the procedure of generating contents with GenAI;
{Act.4.1} Activities intended to set or aligning with a design goal;
{Act.4.2} Acactivities intended to introduce students in familiarising, collecting or creating prompts (data);
{Act.4.3} Activities intended to provide a step-by-step tool tutorial, such as Midjourney, Dall-E;
{Act.5} Activities intended to allow students to individually generate contents with GenAI for a design goal;
{Act.6} Activities intended to provide students with a structured template for process and results documentation;
{Act.7} Activities intended to make students explaining and questioning the process;

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3.3 Sub-module 1: Getting Familiar with Data Visualization (DV)

This sub-module teaches students to apply Computer Vision (CV) algorithms to analyse and visualise data [DV-Obj]. Students develop procedural knowledge through practical guided activities (e.g. Midjourney step-by-step tool tutorials) and they are asked to apply the acquired procedure to a familiar task (e.g. creating an infographic). Students are first provided with a basic literacy of DV algorithms and data visualisation (DV) techniques including technical vocabulary (DV-Act1) and primary functionalities (DV-Act2). This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the disciplines (CV and DV) and their interrelationship in analysing and visualising complex phenomena (Omena, 2021). Once basic literacy is introduced, specific case studies are provided to contextualise the application of CV within the design research (DV-Act3).

At this point, the teaching moves from a theoretical to a procedural level. In this direction, students are first introduced to a guided tutorial (DV-Act4.3) on how to analyse data with CV algorithms (e.g. Meme Spector) and visualise the results (e.g. Gephi) starting from setting a research question (DV-Act4.1) and collecting and organising data related to the phenomena (DV-Act4.2). Then, to familiarise themselves with the procedure, students individually apply what they learned by investigating the dataset to answer the research question (DV-Act5). Lastly, moving the focus to meta-cognitive knowledge, students are asked to analyse and document the process by breaking down the steps and selecting the relevant results (DV-Act6). This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others (DV-Act7).

Educational Objective {DV-Obj} :

Students will learn to apply Computer Vision algorithms for analyze and visualize data.

Sub-module 1: Getting familiar w/ DV	The cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
(a) Factual Knowledge		{DV-Act.1}				
(b) Conceptual Knowledge		{DV-Act.2} {DV-Act.3}				
(c) Procedural Knowledge		{DV-Act.4} {DV-Act.4.1, 4.2, 4.3}	{DV-Obj} {DV-Act.5}			
(d) Meta Knowledge				{DV-Act.6}	{DV-Act.7}	

{DV-Act1} Activities intended to provide students with vocabulary knowledge of Computer Vision;
{DV-Act2} Activities intended to provide students with a basic functionality of DV Algorithms and DV technniques;
{DV-Act3} Activities intended to provide students with practical examples of computer vision applied to design research;
{DV-Act4} Activities intended to explain the procedure of analyzing data with computer vision algorithms;
{DV-Act4.1} Activities intended to introduce students in familiarising, collecting or creating data;
{DV-Act4.2} Activities intended to set or aligning with a research question;
{DV-Act4.3} Activities intended to provide a step-by-step tool tutorial, such as Memespecotor, Gephi;
{DV-Act5} Activities intended to allow students to individually analyzes data with computer vision for a research goal:
{DV-Act6} Activities intended to provide students with a structured template for process and results documentation;
{DV-Act7} Activities intended to make students explaining and questioning the process.

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